VELOOP REWARDS - Production Event Tracking Plan

Data Analyst Internship, Task 01 | Work Package F

Authenticity notice: Every event below is **Proposed** or **To Be Verified**. No event is claimed to exist in VELOOP, and Implementation Status is 'To be confirmed by Engineering' for all. No simulated or real VELOOP statistics appear here.

1. Summary

- **42 events** across Account, Tasks, Rewards, Daily and Engagement, Referrals, Wallet, Withdrawals, Support and third-party earning; 9 are rated Critical.
- **All 32 events** in the brief's minimum list are covered (checked by formula in the spreadsheet).
- **Every event carries the 16 required fields** from the brief, plus a Priority. The full catalogue is in 08_Event_Tracking_Plan.xlsx; this report shows the naming standard, common properties, an overview and required properties for engineering review.

2. Naming standard

Rule	Standard	Example
Pattern	object_action: the thing, then what happened to it, in snake_case	task_completed, reward_credited, user_registered
Tense	Action in the past tense, because an event records something that has already happened	withdrawal_completed, not withdrawal_complete
State-entry exceptions	Two events keep the state word from the brief's minimum list: reward_pending and withdrawal_processing	Documented exceptions, nothing else uses this form
Case and characters	Lowercase letters, digits and underscores only; no spaces, no abbreviations unless standard	wallet_transaction_created
Feature in a property, not the name	Use a generic event with a feature or source_feature property when behaviour is the same across features	reward_credited with source_feature = spin, not spin_reward_credited
Feature-specific objects	Allowed when the object is itself a feature object	spin_started, mission_completed, mining_claimed
Outcome vs interaction	Interactions can come from the frontend (task_opened); outcomes must come from the backend (reward_credited, task_completed)	Prevents counting what the UI showed as what actually happened
Stability	Released event names are not renamed. A change is a new version or a new event, recorded in the change log	See Work Package Q
No personal data	No personal data in event names or properties	See Work Package B excluded data

Why object_action in the past tense:

- Matches the brief's minimum event list, so engineering sees familiar names.
- Short and readable. domain_object_action (e.g. wallet_withdrawal_requested) adds length without adding meaning, because the object name already identifies the domain.
- Events sort and group by object in tools, so all wallet_* or withdrawal_* events appear together.
- Past tense makes it clear that an event is a fact, not a command, which supports the rule that outcomes come from the backend.
- Generic events with a feature property keep the catalogue small, avoiding a separate reward event per feature.

3. Common properties on every event

These are in addition to each event's own properties, so every event can be deduplicated, ordered and traced to a user.

Property	Type	Required?	Meaning
event_id	string	Yes	Unique key for this event occurrence; a retry reuses the same id
event_name	string	Yes	Name from the catalogue, following the naming standard
event_timestamp	timestamp (UTC)	Yes	When the event happened, ISO 8601 UTC
received_timestamp	timestamp (UTC)	Yes (added on receipt)	When the analytics pipeline received it; detects delay and clock problems
user_id	string	Yes	Stable user key; never email, phone or name
schema_version	string	Yes	Version of the event definition, so changes stay traceable
platform	string (enum)	Yes	android, ios or web (list to be agreed)
app_version	string	Yes	Version of the app that emitted the event
source	string (enum)	Yes	frontend, backend or provider
session_id	string	Optional	Present only if session tracking is adopted

4. Event overview

Event	Feature	Trigger	Source	Priority	Status
user_registered	Account	Account creation succeeds	Backend	Critical	To Be Verified
user_logged_in	Account	Login succeeds	Backend	High	Proposed
user_logged_out	Account	User chooses logout (where logout exists)	Frontend	Low	Proposed
onboarding_started	Account	First onboarding screen shown	Frontend	High	Proposed
onboarding_completed	Account	Final onboarding step done	Backend	High	Proposed
account_status_changed	Account	account_status updated by user, system or operations	Backend	High	To Be Verified
task_viewed	Tasks	Task card visible on screen	Frontend	Medium	Proposed
task_opened	Tasks	User opens task detail	Frontend	Medium	Proposed
task_started	Tasks	Task attempt record created	Backend	Critical	Proposed
task_completed	Tasks	Completion has passed the applicable validation	Backend	Critical	Proposed
task_failed	Tasks	Attempt ends without validated completion	Backend	High	Proposed
reward_pending	Rewards	Reward created but not yet usable	Backend	High	Proposed
reward_credited	Rewards	Wallet balance updated with the reward	Backend	Critical	Proposed
reward_reversed	Rewards	A credited reward is taken back	Backend	Critical	Proposed
reward_corrected	Rewards	Reward amount changed after crediting	Backend	Medium	Proposed
reward_rejected	Rewards	Pending reward will never be credited (if this state exists)	Backend	Medium	To Be Verified
daily_reward_claimed	Daily & Engagement	Daily reward successfully claimed	Backend	High	Proposed
streak_updated	Daily & Engagement	Streak incremented or reset	Backend	Medium	Proposed
mission_started	Daily & Engagement	Mission attempt begins	Backend	Medium	Proposed
mission_completed	Daily & Engagement	System marks mission complete	Backend	Medium	Proposed
badge_earned	Daily & Engagement	Badge awarded	Backend	Low	Proposed
spin_started	Daily & Engagement	User starts a spin	Frontend	Medium	Proposed
spin_completed	Daily & Engagement	Spin outcome decided	Backend	Medium	Proposed
mining_started	Daily & Engagement	Mining cycle begins	Backend	Medium	To Be Verified
mining_claimed	Daily & Engagement	User claims mining result	Backend	Medium	To Be Verified
feature_opened	Daily & Engagement	User opens a feature screen (name to be agreed)	Frontend	Medium	To Be Verified
referral_created	Referrals	Referral code or link created	Backend	High	Proposed
referral_registered	Referrals	Referred user completes sign-up	Backend	High	Proposed
referral_activated	Referrals	Referred user meets activation rule	Backend	High	To Be Verified
referral_converted	Referrals	Referral meets success condition	Backend	High	Proposed

Event	Feature	Trigger	Source	Priority	Status
wallet_transaction_created	Wallet	Ledger entry created	Backend	Critical	To Be Verified
wallet_transaction_completed	Wallet	Ledger entry becomes final	Backend	High	To Be Verified
wallet_transaction_reversed	Wallet	Reversal entry created	Backend	Critical	To Be Verified
wallet_exchange_completed	Wallet	Exchange of units finalised	Backend	Medium	To Be Verified
withdrawal_requested	Withdrawals	Withdrawal request created	Backend	Critical	To Be Verified
withdrawal_processing	Withdrawals	Request enters review or processing	Backend	High	To Be Verified
withdrawal_completed	Withdrawals	Payout finished	Backend	Critical	To Be Verified
withdrawal_rejected	Withdrawals	Request refused	Backend	High	To Be Verified
withdrawal_reversed	Withdrawals	Completed payout undone or returned	Backend	High	To Be Verified
withdrawal_cancelled	Withdrawals	User cancels before processing (if allowed)	Backend	Low	To Be Verified
support_request_created	Support	User submits a support request (name to be agreed)	Backend	Low	To Be Verified
provider_callback_received	Third-party earning	External provider reports a result (name to be agreed)	Provider	High	To Be Verified

5. Required and optional properties with validation

Data types for each property are in the spreadsheet. 'Plus common properties' applies to every row.

Event	Required Properties	Optional Properties	Validation
user_registered	registration_source	registration_method, referral_code_used, campaign_id, country_code	registration_source in approved list; exactly one event per user_id; timestamp not in future
user_logged_in		attempt_number	user_id must exist; timestamp not before registration
user_logged_out			Not before a prior user_logged_in; may be absent for timeouts
onboarding_started	onboarding_step		onboarding_step in current flow list; once per user unless flow restarts
onboarding_completed	onboarding_step	step_completed_count	Must follow onboarding_started; once per user
account_status_changed	previous_status, new_status, status_change_actor	reason_category	previous_status differs from new_status; values in approved status list
task_viewed	task_id, task_type, feature	screen_name	Only when visible, not when merely loaded; sample if volume is high
task_opened	task_id, task_type, feature		task_id exists in catalogue
task_started	task_activity_id, task_id, task_type, feature	provider_name, provider_reference_id, attempt_number	task_activity_id unique; task_id exists; one event per attempt
task_completed	task_activity_id, task_id, task_type, reward_amount, reward_unit, validation_status	reward_id, provider_reference_id	validation_status = passed; exactly one per task_activity_id; reward_unit in approved list; reward_amount not negative
task_failed	task_activity_id, task_id, failure_category	failure_detail_code, provider_name, provider_reference_id	failure_category in approved list; cannot follow task_completed for the same attempt
reward_pending	reward_id, reward_amount, reward_unit, source_feature, pending_reason	task_activity_id	reward_id unique; reward_amount greater than 0
reward_credited	reward_id, reward_amount, reward_unit, source_feature, wallet_transaction_id	task_activity_id	Exactly one per reward_id; wallet_transaction_id must exist; amount greater than 0
reward_reversed	reward_id, original_reward_id, reward_amount, reward_unit, reason_category, wallet_transaction_id		original_reward_id must refer to a credited reward; no second reversal of the same reward
reward_corrected	reward_id, old_amount, new_amount, reward_unit, correction_reason	wallet_transaction_id	old_amount differs from new_amount; reward must exist
reward_rejected	reward_id, reason_category		Must follow reward_pending; cannot follow reward_credited
daily_reward_claimed	claim_date, streak_day, reward_amount, reward_unit	reward_id	One per user per claim_date; reward_amount not negative
streak_updated	streak_count, previous_streak_count, change_type		change_type in (increment, reset); counts consistent with change_type
mission_started	mission_id, mission_type		mission_id exists
mission_completed	mission_id, mission_type	reward_amount, reward_unit, progress_value	Must follow mission_started; once per mission attempt
badge_earned	badge_id		One per user per badge_id
spin_started	spin_id		spin_id unique
spin_completed	spin_id, outcome_category	reward_amount, reward_unit	Must follow spin_started; one per spin_id

Event	Required Properties	Optional Properties	Validation
mining_started	mining_cycle_id		mining_cycle_id unique
mining_claimed	mining_cycle_id, reward_amount, reward_unit	reward_id	Must follow mining_started; once per cycle
feature_opened	feature_name	screen_name	feature_name in approved list; used only for features without their own start event
referral_created	referral_id, referring_user_id	referral_code_used, campaign_id	referral_id unique; referring_user_id must exist
referral_registered	referral_id, referring_user_id, referred_user_id		referred_user_id differs from referring_user_id; follows referral_created
referral_activated	referral_id, referred_user_id, activation_rule_version		Follows referral_registered; once per referral_id
referral_converted	referral_id, referring_user_id, referred_user_id	reward_id, wallet_transaction_id	Follows referral_activated unless rule says otherwise; once per referral_id
wallet_transaction_created	wallet_transaction_id, transaction_type, direction, amount, unit, reference_id, source_feature	related_transaction_id, exchange_group_id	wallet_transaction_id unique; amount greater than 0; direction agrees with transaction_type
wallet_transaction_completed	wallet_transaction_id, transaction_type, direction, amount, unit, transaction_status		Follows wallet_transaction_created; once per transaction
wallet_transaction_reversed	wallet_transaction_id, related_transaction_id, amount, unit, reason_category		related_transaction_id exists and is not itself a reversal
wallet_exchange_completed	exchange_group_id, from_unit, to_unit, from_amount, to_amount	exchange_rate_applied	One debit and one credit per exchange_group_id
withdrawal_requested	withdrawal_id, requested_amount, requested_unit, method	wallet_transaction_id	withdrawal_id unique; requested_amount greater than 0; method in approved list
withdrawal_processing	withdrawal_id, method	provider_reference_id	Follows withdrawal_requested; once per withdrawal
withdrawal_completed	withdrawal_id, method, completed_amount, requested_unit, wallet_transaction_id	provider_reference_id	Follows withdrawal_processing; one linked debit; completed_amount not above requested unless a fee rule exists
withdrawal_rejected	withdrawal_id, method, reason_category		reason_category required; cannot follow withdrawal_completed
withdrawal_reversed	withdrawal_id, method, reason_category, wallet_transaction_id	provider_reference_id	Must follow withdrawal_completed
withdrawal_cancelled	withdrawal_id, reason_category		Must precede withdrawal_processing
support_request_created	ticket_id, issue_category	article_id	issue_category in approved list; no free text in analytics
provider_callback_received	provider_name, provider_reference_id, provider_status, callback_received_at	provider_event_time, task_activity_id	provider_reference_id present; stored separately from VELOOP status

6. Open questions for verification

- Which of these events are emitted today, and by which component? (A001, A011)
- For each event, is the frontend or backend the right emitter? Rule proposed: interactions from the frontend, outcomes from the backend.
- What idempotency design prevents a retried request from producing two events? (A005)
- Is task_viewed volume acceptable, or should it be sampled?
- Do features outside earning need feature_opened, or do they have their own events?
- Which statuses, units, methods and categories form the approved lists used in validation? (A002, A006)

Next: Work Package G - VELOOP Data Dictionary (75+ meaningful fields).